

Problem 1. (2019 XXII All-Ukrainian Tournament of Young Mathematicians P17) n points are marked on the board points that are vertices of the regular n -gon. One of the points is a chip. Two players take turns moving it to the other marked point and at the same time draw a segment that connects them. If two points already connected by a segment, such a move is prohibited. A player who can't make a move, lose. Which of the players can guarantee victory?

Ans. First player always wins. We will split the vertices into two sets: $\{1, 2\}$ and $S = \{3, \dots, n\}$. In the first move, player 1 makes edge $(1, 2)$, for whatever moves player 2 moves, say $(2, k)$ for $k \in S$, player 1 makes move $(k, 1)$. With this process, the game will always be played on a complete bipartite graph $K_{2, n-2}$. Since there's even number of edges, and player 2 makes the first move in the bipartite graph, we have player 1 always win.

Problem 2. (2021 Ukraine NMO P9) Alexey and Bogdan play a game with two piles of stones. In the beginning, one of the piles contains 2021 stones, and the second is empty. In one move, each of the guys has to pick up an even number of stones (more than zero) from an arbitrary pile, then transfer half of the stones taken to another pile, and the other half - to remove from the game. Loses the one who cannot make a move. Who will win this game if both strive to win, and Bogdan begins?

Ans. First player will win.

Claim: A player who made piles like n and $n - 1$, has a winning strategy.

Proof: The next player will transfer the piles to $(n - 2k, n - 1 + k)$ or $(n + k, n - 1 - 2k)$, in either case, we can reduce it back to $(n - k, n - k - 1) = (n', n' - 1)$. Since the ending state is $(0, 1)$, which is in the form of $(n, n - 1)$.

Since Alexey can transform $(2021, 0) \rightarrow (673, 674)$, he wins. $\square$

Problem 3. (SAMO 2017 Q4) Andile and Zandre play a game on a 2017×2017 board. At the beginning, Andile declares some of the squares forbidden, meaning the nothing may be placed on such a square. After that, they take turns to place coins on the board, with Zandre placing the first coin. It is not allowed to place a coin on a forbidden square or in the same row or column where another coin has already been placed. The player who places the last coin wins the game. What is the least number of squares Andile needs to declare as forbidden at the beginning to ensure a win? (Assume that both players use an optimal strategy.)

Ans. The answer is 2017. For 2017, Andile can just declare an entire row forbidden. Now there are 2016 rows and 2017 columns, and each move will reduce the number of available rows and columns each by 1. As there are more columns than rows, the game ends when we run out of rows. Since 2016 is even, Andile wins.

Now we show that 2016 is not enough. We will prove the following statement by induction.

Claim: On a $(2n + 1) \times (2n + 1)$ board, if at most $2n$ squares are forbidden then Zandre wins.

Proof: The base case $n = 0$ is trivial. Now suppose we have a $(2k + 1) \times (2k + 1)$ board with at most $2k$ forbidden squares ($k \geq 1$).

- If there are at most one forbidden square, Zandre chooses some other square in its row. Now whatever square Andile chooses, we're left with 0 forbidden squares on a $(2k - 1) \times (2k - 1)$ board.
- Else there are at least 2 forbidden squares. If some two of them are in the same row or column, Zandre chooses a non-forbidden square in that row or column (which exists since $2k < 2k + 1$).
- Else consider two forbidden squares X, Y . Zandre choose the square Z on the same row as X and same column as Y . Now whatever square Andile chooses, we're left with at most $2k - 2$ forbidden squares on a $(2k - 1) \times (2k - 1)$ board. $\square$

Hence Andile needs at least 2017 forbidden squares.

Problem 4. Ana and Bruno decide to play a game with the following rules. Ana has cards $1, 3, 5, 7, \dots, 2n - 1$, and Bruno has cards $2, 4, 6, 8, \dots, 2n$. During the first turn and all odd turns afterwards, Bruno chooses one of his cards first and reveals it to Ana, and Ana chooses one of her cards second. Whoever's card is higher gains a point. During the second turn and all even turns afterwards, Ana chooses one of her cards first and reveals it to Bruno, and Bruno chooses one of his cards second. Similarly, whoever's card is higher gains a point. During each turn, neither player can use a card they have already used on a previous turn. The game ends when all

cards have been used after n turns. Determine the highest number of points Ana can earn.

Ans. We claim that Anna can gain a maximum of $\lfloor \frac{n}{2} \rfloor$ points. We prove by induction that this maximum score is possible.

Base case: $n = 1$ is trivial. For $n = 2$, if Bruno plays 4, Anna plays 1 and she wins the next point. If Bruno plays 2 first, Anna plays 3 and wins the first point and loses the second.

Induction: Now assume the maximum score is possible for $n - 2$. For n , if Bruno plays $2n$, Anna plays 1, and the next turn no matter what Bruno plays, Anna plays the number above. If Bruno plays anything lower than $2n$ first, Anna plays the next number higher and thus wins that point. Anna then plays 1 and loses the second point. At this point, when Anna and Bruno's cards are considered together from largest to smallest, Bruno has the highest card and every other card after that, and Anna has the second highest card and every other card after that, so the case is identical to the $n - 2$ case and since Anna has one more point going into that case, the maximum is always possible by induction.

We now prove that no higher score is possible. This is equivalent to proving that Bruno can always gain $\lceil \frac{n}{2} \rceil$ points. Assume Bruno plays $2n$ on his first move, guaranteeing him that point. Then Anna optimally plays 1; since she will lose the point anyways, it is obviously optimal for her to maximize the cards she has left. But here, the roles of Bruno and Anna are reversed: Anna, having the highest card and every other card after that, plays first this turn, and Bruno has the second highest card and every other card after that. So this case is identical to the previous case with Anna. Thus Bruno can gain another $\lfloor \frac{n-1}{2} \rfloor$ points, giving him $\lfloor \frac{n+1}{2} \rfloor$ points in total, equal to $\lceil \frac{n}{2} \rceil$ points as n is an integer. So thus no higher score for Anna is possible and the result is proved.

Problem 5. (2019 Central American and Caribbean Mathematical Olympiad P2) We have a regular polygon P with 2019 vertices, and in each vertex there is a coin. Two players Azul and Rojo take turns alternately, beginning with Azul, in the following way: first, Azul chooses a triangle with vertices in P and colors its interior with blue, then Rojo selects a triangle with vertices in P and colors its interior with red, so that the triangles formed in each move don't intersect internally the previous colored triangles. They continue playing until it's not possible to choose another triangle to be colored. Then, a player wins the coin of a vertex if he colored the greater quantity of triangles incident to that vertex (if the quantities of triangles colored with blue or red incident to the vertex are the same, then no one wins that coin). The player with the greater quantity of coins wins the game. Find a winning strategy for one of the players. (Two triangles can share vertices or sides.)

Ans. Azul wins. Azul starts by picking triangle $A_1A_{1010}A_{1011}$, the points are split into two sets $S = \{A_2, A_3, \dots, A_{1009}\}$ and $\bar{S} = \{A_{1012}, A_{1013}, \dots, A_{2019}\}$. It is clear that any other triangle must have all its vertices in one of these two sets. On the successive rounds Azul just mimic the previous triangle chosen by Rojo in the complementary set. We can then observe the following:

1. Azul and Rojo are even on $S \cup \bar{S}$.
2. Azul and Bojo are even on $\{A_{1010}, A_{1011}\}$.
3. A_1 is for Azul.

Hence Azul will have at least one extra coin than Rojo (A_1 is always for Azul), and hence Azul wins.

Problem 6. (Kosovo TST 2019 P1) There are 2019 cards in a box. Each card has a number written on one of its sides and a letter on the other side. Amy and Ben play the following game: in the beginning Amy takes all the cards, places them on a line and then she flips as many cards as she wishes. Each time Ben touches a card he has to flip it and its neighboring cards. Ben is allowed to have as many as 2019 touches. Ben wins if all the cards are on the numbers' side, otherwise Amy wins. Determine who has a winning strategy.

Ans. We will show that Beto has the winning strategy. First of all ignore the condition that he can make at most 2022 moves. We will allow an arbitrary number of moves, and then if we applied k moves on a certain location, we replace k with $k \pmod{2}$ and become safe. The main portion of the proof that follows hinges on $2022 \equiv 0 \pmod{3}$. Now we run a linear scan on all indices from left to right. (let us number them from 1 to 2022 for convenience) If the current card is colored White, just pass through it, otherwise flip the card next to it. (we stop when we reach the last card, and do not apply this move there) This ensures that the first 2021 cards are White. If the last one is also white, we are done. So assume that it is Black. If we apply the given type of on consecutive cards, then we immediately notice that we can swap indices i and $i + 3$ at the same time. Inductively, this also yields that it is possible to swap i and j with $i \equiv j \pmod{3}$ without affecting rest of the board. In particular we can do this on 3 and 2022. So now location 3 is black, all others are white. Apply the

move on location 2 and then location 1. This completes the algorithm.

Remark: Show that the conclusion holds for any general $n \geq 3$ with $n \not\equiv 2 \pmod{3}$ instead of 2022. What happens if $n \equiv 2 \pmod{3}$?

Problem 7. (Brazilian Undergrad MO 2020 P5) Let N a positive integer. In a spaceship there are $2 \cdot N$ people, and each two of them are friends or foes (both relationships are symmetric). Two aliens play a game as follows:

1. The first alien chooses any person as she wishes.
2. Thenceforth, alternately, each alien chooses one person not chosen before such that the person chosen on each turn be a friend of the person chosen on the previous turn.
3. The alien that can't play in her turn loses.

Prove that second player has a winning strategy if, and only if, the $2 \cdot N$ people can be divided in N pairs in such a way that two people in the same pair are friends.

Ans. ($\Leftarrow$) If the $2N$ people can be divided into N pairs, then the second player has a winning strategy. If the people can be paired $(a_1, b_1), (a_2, b_2), \dots, (a_N, b_N)$ such that a_i and b_i are friends, then the strategy of the second player is to choose the pair of the person selected by the first player in the previous turn.

($\Rightarrow$) If the second player has a winning strategy, then the $2N$ people can be divided into N pairs. We prove its contrapositive, i.e. if the $2N$ people can't be divided into N such pairs, then the second player doesn't have a winning strategy. Consider the graph G with the people as the vertices, and two people are connected by an edge if they are friends. Now, if the people can't be divided into N pairs of friends, the maximum matching of G is less than N . We pair the vertices according to this maximum matching, and we will show a winning strategy for the first player (hence the second player loses and doesn't have a winning strategy).

In the first turn, the first player chooses a person without a pair (call this person v_1). Then, the second player must choose a friend of v_1 . Any friend of v_1 must be contained in a pair, otherwise, this person can be paired with v_1 and creates an additional pair, which contradicts the fact that we pair the people according to the maximum matching. Let the second player choose v_2 . Then, in the next turn, the first player chooses the pair of v_2 . So the strategy of the first player is to choose the pair of the person selected at the previous turn by the second player. If the second player can always move, eventually at some point, the second player chooses v_{2k} , which is not contained in any pairs. Consider the following path:

$$v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_{2k}.$$

We know that v_1 and v_{2k} both are not contained in any pairs, and $(v_2, v_3), (v_4, v_5), \dots, (v_{2k-2}, v_{2k-1})$ are pairs in the maximum matching. But since v_i and v_{i+1} are friends, then we can create an additional pair by shifting the pairings as follows: $(v_1, v_2), (v_3, v_4), \dots, (v_{2k-1}, v_{2k})$. This is a contradiction. Therefore, at some point the second player couldn't move, and the first player wins.

Problem 8. (Putnam 2013 B6) Let $n \geq 1$ be an odd integer. Alice and Bob play the following game, taking alternating turns, with Alice playing first. The playing area consists of n spaces, arranged in a line. Initially all spaces are empty. At each turn, a player either

1. places a stone in an empty space, or
2. removes a stone from a nonempty space s , places a stone in the nearest empty space to the left of s (if such a space exists), and places a stone in the nearest empty space to the right of s (if such a space exists).

Furthermore, a move is permitted only if the resulting position has not occurred previously in the game. A player loses if he or she is unable to move. Assuming that both players play optimally throughout the game, what moves may Alice make on her first turn?

Ans. We show that the only winning first move for Alice is to place a stone in the central space. We start with some terminology.

By a *block* of stones, we mean a (possibly empty) sequence of stones occupying consecutive spaces. By the *extremal blocks*, we mean the (possibly empty) maximal blocks adjacent to the left and right ends of the playing area.

We refer to a legal move consisting of placing a stone in an empty space as a move of *type 1*, and any other legal move as being of *type 2*. For $i = 0, \dots, n$, let P_i be the collection of positions containing i stones. Define the *end zone* as the union $Z = P_{n-1} \cup P_n$. In this language, we make the following observations.

- Any move of type 1 from P_i ends in P_{i+1} .
- Any move of type 2 from P_n ends in P_{n-1} .
- For $i < n$, any move of type 2 from P_i ends in $P_i \cup P_{i+1}$.
- At this point, we see that the number of stones cannot decrease until we reach the end zone.
- For $i < n - 1$, if we start at a position in P_i where the extremal blocks have length a, b , then the only possible moves to P_i decrease one of a, b while leaving the other unchanged (because they are separated by at least two empty spaces). In particular, no repetition is possible within P_i , so the number of stones must eventually increase to $i + 1$.
- From any position in the end zone, the legal moves are precisely to the other positions in the end zone which have not previously occurred. Consequently, after the first move into the end zone, the rest of the game consists of enumerating all positions in the end zone in some order.
- At this point, we may change the rules without affecting the outcome by eliminating the rule on repetitions and declaring that the first player to move into the end zone loses (because $\#Z = n + 1$ is even).

To determine who wins in each position, number the spaces of the board $1, \dots, n$ from left to right. Define the *weight* of a position to be the sum of the labels of the occupied spaces, reduced modulo $n + 1$. For any given position outside of the end zone, for each $s = 1, \dots, n$ there is a unique move that adds s to the weight: if s is empty that a move of type 1 there does the job. Otherwise, s inhabits a block running from $i + 1$ to $j - 1$ with i and j empty (or equal to 0 or $n + 1$), so the type 2 move at $i + j - s$ (which belongs to the same block) does the job.

We now verify that a position of weight s outside of the end zone is a win for the player to move if and only if $s \neq (n + 1)/2$. We check this for positions in P_i for $i = n - 2, \dots, 0$ by descending induction. For positions in P_{n-2} , the only safe moves are in the extremal blocks; we may thus analyze these positions as two-pile Nim with pile sizes equal to the lengths of the extremal blocks. In particular, a position is a win for the player to move if and only if the extremal blocks are unequal, in which case the winning move is to equalize the blocks. In other words, a position is a win for the player to move unless the empty spaces are at s and $n + 1 - s$ for some $s \in \{1, \dots, (n - 1)/2\}$, and indeed these are precisely the positions for which the weight equals $(1 + \dots + n) - (n + 1) \equiv (n + 1)/2 \pmod{n + 1}$. Given the analysis of positions in P_{i+1} for some i , it is clear that if a position in P_i has weight $s \neq (n + 1)/2$, there is a winning move of weight t where $s + t \equiv (n + 1)/2 \pmod{n}$, whereas if $s = (n + 1)/2$ then no move leads to a winning position.

It thus follows that the unique winning move for Alice at her first turn is to move at the central space, as claimed.